

StoreDB — SQL + Power BI Analytics Project

A retail analytics project demonstrating SQL data analysis, data cleaning, DAX, data modeling, and Power BI dashboard development.

Project Overview

StoreDB is a retail-style relational database containing customers, orders, and order-item transaction data. MySQL is used for data preparation and analysis, while Power BI is used for interactive business reporting.

Database Structure

Table	Records	Purpose
Customers	250	Customer details and city information
Orders	650	Order dates and customer relationships
Order_Items	1,150	Products, quantities and unit prices

Data note: The supplied SQL file defines 1,150 Order_Items records and ends with a verification COUNT(*) query. This README therefore reports 1,150 as the documented Order_Items count.

SQL Analysis

- Database and relational table creation using MySQL.
- Verification of table row counts and data integrity.
- Duplicate-line-item analysis using ROW_NUMBER().
- Orphaned-order checks using LEFT JOIN.
- Quantity and unit-price quality checks.
- Executive KPIs: total orders, units, revenue, and average order value.
- Customer ranking by revenue and order volume.
- Inactive-customer analysis and customer spend tiers.
- Top-product analysis by revenue and units sold.
- Geographic market analysis by city.
- Top-spending customer per city using DENSE_RANK().
- Monthly order volume and revenue analysis.
- Creation of the BI reporting view: vw_customer_order_summary.

Power BI Dashboard

Page 1 — Executive Sales Dashboard

KPIs: Total Revenue, Total Orders, Total Units, Total Customers, and Average Order Value. Visuals include monthly revenue, revenue by city, top products by revenue, and units sold by product.

Page 2 — Customer Analytics

Customer KPIs, top customers by revenue, customer performance table, and customer revenue by city.

Page 3 — Product Analysis

Product revenue, units sold, product performance, and top products by average unit price.

DAX Measures

```
Revenue = 'storedb order_items'[quantity] * 'storedb order_items'[unit_price]
```

```
Total Revenue = SUM('storedb order_items'[Revenue])
```

```
Total Orders = DISTINCTCOUNT('storedb orders'[order_id])
```

```
Total Units = SUM('storedb order_items'[quantity])
```

```
Total Customers = DISTINCTCOUNT('storedb customers'[customer_id])
```

```
Average Order Value = DIVIDE([Total Revenue], [Total Orders])
```

Tools & Technologies

MySQL 8.0+ • MySQL Workbench • SQL • Power BI • DAX • Data Modeling • GitHub

Repository Structure

```
StoreDB-Analytics/  
■■■ StoreDB.sql  
■■■ StoreDB_PowerBI_Dashboard.pbix  
■■■ README.md  
■■■ Screenshots/  
■■■ Executive-Sales-Dashboard.png  
■■■ Customer-Analytics.png  
■■■ Product-Analysis.png
```

Project Purpose

This project demonstrates practical analytics skills including SQL querying, relational data analysis, KPI development, DAX calculations, Power BI visualization, data modeling, and business reporting.

Author

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